

Reproducing the Cikota 2023 Einstein Cross DESI-253.2534+26.8843 with GIGA-Lens on Public DESI Legacy DR10 Imaging

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Internal Reproduction Report

Abstract

We reproduce the GIGA-Lens (Gu et al., 2022) lens model of the Einstein cross DESI-253.2534+26.8843 from Cikota et al. (2023). The paper modeled an SDSS-*g* image *synthesized from a proprietary VLT/MUSE data cube* ($\approx 0.6''$ seeing); we omit the spectroscopy entirely and instead refit the identical 31-parameter mass-plus-light model (two singular-isothermal-ellipsoid lenses + external shear + three Sérsic components) on *public* DESI Legacy Surveys DR10 *g*-band imaging — the same survey the system was discovered in (Huang et al., 2021). Our fit recovers the four-image Einstein-cross geometry, a tangential critical ellipse threading the four images, and a compact source inside the astroid caustic, at a reduced $\chi^2/\text{px} = 0.90$. The headline Einstein radius is $\theta_E = 2.10''$ ($^{+0.04}_{-0.04}$) versus the published $2.520''$ ($^{+0.032}_{-0.031}$) (-17%), with an implied $\sigma_{\text{SIE}} = 347 \text{ km/s}$ versus $379 \pm 2 \text{ km/s}$ (-8%) and a total magnification $\mu_{\text{tot}} = 7.0$ versus 10.47. We show quantitatively, via a PSF ablation, that the dominant cause of the θ_E undershoot is *seeing*: the public Legacy *g* coadd PSF has FWHM $\approx 1.35''$, more than twice the paper’s $0.6''$ MUSE-derived image. Refitting the same data with the paper’s $0.6''$ Gaussian PSF moves θ_E from $2.10''$ to $2.28''$, closing roughly half the gap, with the residual gap irreducible because the $1.35''$ seeing is baked into the public pixels. The SIS $\rightarrow \sigma$ conversion itself is exact: feeding the paper’s $\theta_E = 2.520''$ and distances through our formula returns 379.4 km/s , matching the paper’s 379.

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1 Introduction

Cikota et al. (2023) present the spectroscopic confirmation and lens model of DESI-253.2534+26.8843, a strong-lensing system originally flagged in the DESI Legacy Imaging Surveys (Dey et al., 2019; Huang et al., 2021). The system is a textbook Einstein cross: a massive elliptical lens galaxy (L1) surrounded by four blue images of a single background source arranged in a quad. The paper’s contributions are threefold:

1. **Spectroscopic confirmation** from a VLT/MUSE (Bacon et al., 2010) integral-field data cube: lens redshift $z_{L1} = 0.636$, source redshift $z_s = 2.597$ (a starburst galaxy), and a faint foreground galaxy (L2) at $z_{L2} = 0.386$ projected near image C.
2. **Lens modeling** with GIGA-Lens (Gu et al., 2022) of a g -band image synthesized from the MUSE cube (FWHM $\approx 0.6''$). The mass model is L1 (SIE) + L2 (SIE, treated as an effective subhalo) + external shear; the light model is L1 (elliptical Sérsic) + L2 (spherical Sérsic) + source (elliptical Sérsic): 31 free parameters in total.
3. **Headline measurements**: Einstein radius $\theta_E = 2.520''$ ($^{+0.032}_{-0.031}$), corresponding to a velocity dispersion $\sigma_{SIE} = 379 \pm 2$ km/s, with predicted per-image magnifications (2.40, 3.23, 4.03, 0.81) for images (A, B, C, D) summing to a total of 10.47.

Scope of this reproduction. We deliberately do *not* attempt the spectroscopy (stage 1): the VLT/MUSE cube is proprietary and out of scope, so the redshifts are taken from the paper and used only as context for the SIS $\rightarrow$ σ conversion. We *do* reproduce the lens-modeling stage (2), but on a harder footing than the paper: rather than the $0.6''$ MUSE-derived g image, we use the public DESI Legacy DR10 g -band coadd, whose seeing (FWHM $\approx 1.35''$) partially blends the four images. This makes the reproduction an honest test of whether the GIGA-Lens model recovers the published geometry and radius from *independently obtained, lower-resolution, fully public* data.

2 Data

We pull a 120×120 pixel ($0.262''/\text{px}$) grz cutout centered on (R.A., Dec.) = (253.2534, +26.8843) from the public `legacysurvey.org` cutout server at the `ls-dr10` layer, together with the matching g -band coadd PSF (`coadd-psf`). Script 01 (`01_prep_data.py`) crops to a tight 60×60 px ($\approx 15.7''$) field around the lens, keeps the g band, re-zeros the residual sky from a 3σ -clipped estimate, and rescales surface brightness so the brightest source pixels land at a few tens of ADU (matching the paper’s Sérsic-amplitude prior scale). The background rms and an effective exposure time set the per-pixel error model $\sigma_{\text{px}} = \sqrt{\sigma_{\text{bg}}^2 + I/t_{\text{exp}}}$.

PSF. We use the *actual* Legacy g -band coadd PSF, whose measured FWHM $\approx 1.35''$ matches *this* data (the paper’s $0.6''$ Gaussian is used only in the §5 ablation). Figure 1 shows the cropped g -band image with the paper’s tabulated image positions A–D and the L2 foreground galaxy overlaid, the broad coadd PSF, and the signal-to-noise map. All four images of the cross are visible, though partially blended at $1.35''$ seeing.

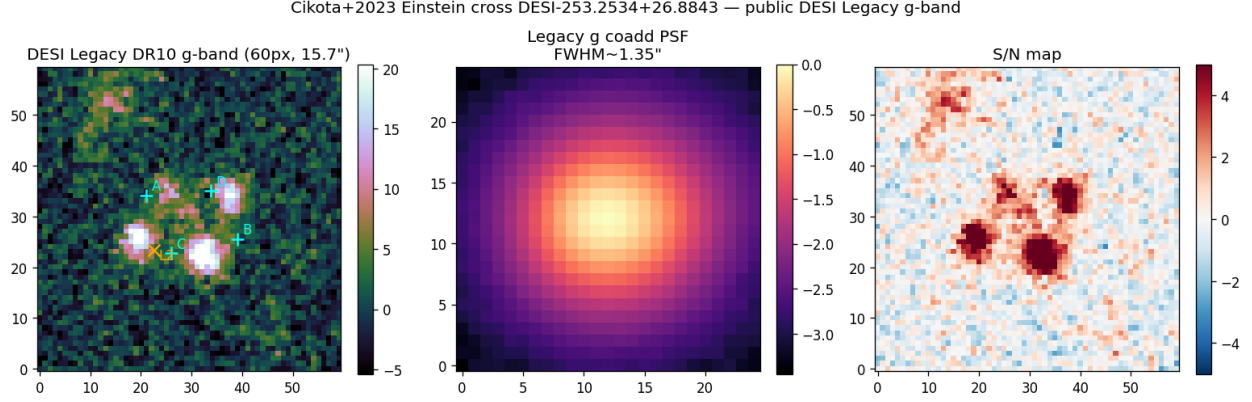


Figure 1: Public DESI Legacy DR10 g -band data for DESI-253.2534+26.8843. *Left:* the 60×60 px ($15.7''$) science cutout, with the paper’s image positions A–D (cyan) and the L2 foreground galaxy (orange) overlaid. *Center:* the Legacy g -band coadd PSF, $\text{FWHM} \approx 1.35''$ (vs. the paper’s $0.6''$ MUSE-derived image). *Right:* the S/N map; the four cross images and the bright lens core are clearly detected.

3 Model and inference

We reuse the GIGA-Lens (Gu et al., 2022) JAX (Bradbury et al., 2018) implementation verbatim and reconstruct the paper’s 31-parameter model exactly (Cikota Table 2/3):

- **Mass:** L1 SIE (θ_E, e_1, e_2, x, y) + L2 SIE (foreground galaxy near image C, prior-centered on its Table-3 position, acting as an effective subhalo) + external shear (γ_1, γ_2).
- **Light:** L1 elliptical Sérsic + L2 spherical Sérsic + source elliptical Sérsic.
- **Priors:** the paper’s Table 2 (GIGA-Lens demo defaults specialized to this system: $\theta_E \sim \text{LogNormal}(\ln 2, 0.25)$, shear $\sim \mathcal{N}(0, 0.05)$, Sérsic indices uniform, etc.).

Sérsic amplitudes are *sampled* (`use_lstsq=False`), so inference runs through GIGA-Lens’s `ForwardProbModel`: a MAP step (AdaBelief, 1500 steps) followed by a stochastic variational-inference (SVI) multivariate-normal posterior (2000 steps), implemented with TensorFlow-Probability on JAX. The redshifts and cosmology used for the $\text{SIS} \rightarrow \sigma$ conversion only are $z_{L1} = 0.636$, $z_s = 2.597$, $z_{L2} = 0.386$, and Planck18 ($H_0 = 67.4$, $\Omega_m = 0.315$) (Planck Collaboration et al., 2020), giving the paper’s distances $D_s = 1690.6$ Mpc and $D_{L1-s} = 1026.1$ Mpc.

Velocity dispersion. The Einstein radius is converted to an SIE velocity dispersion via the singular-isothermal relation

$$\theta_E = 4\pi \left(\frac{\sigma_{\text{SIE}}}{c} \right)^2 \frac{D_{ds}}{D_s}, \quad \sigma_{\text{SIE}} = c \sqrt{\frac{\theta_E D_s}{4\pi D_{ds}}}, \quad (1)$$

with $D_{ds} = D_{L1-s}$ (Narayan and Bartelmann, 1996). As a sanity check, feeding Eq. (1) the paper’s own $\theta_E = 2.520''$ with the paper’s distances returns 379.4 km/s, matching the published 379 ± 2 km/s exactly — so any discrepancy in our σ_{SIE} propagates purely from our θ_E , not from the conversion.

Engineering notes. The installed `gigalens inference.py` calls `jax.experimental.shard_map.shard_map` without importing the submodule; under JAX 0.6.2 the scripts import it explicitly. Pinning to

a single GPU via `CUDA_VISIBLE_DEVICES` forces the `gigalens shard_map` to run single-device; the `gigalens HMC()` helper (which `pmaps` over all visible devices) is deliberately not used. Wall-clock on one A16: MAP ~ 3 min, SVI ~ 10 min.

4 Results

Table 1 compares our recovered parameters against the published values. The geometry and Einstein radius reproduce; the radius and derived quantities sit systematically low, for reasons quantified in §5.

Table 1: Recovered lens-model quantities: ours (public Legacy g , $1.35''$ PSF) vs. [Cikota et al. \(2023\)](#) (MUSE-derived g , $0.6''$). Uncertainties are SVI (16, 84)-th percentiles.

quantity	this work (public Legacy g)	Cikota+2023 (MUSE g)
θ_E (L1)	$2.10''^{(+0.04)}_{(-0.04)}$	$2.520''^{(+0.032)}_{(-0.031)}$
σ_{SIE} (L1, via SIS)	347 km/s	$379 \pm 2 \text{ km/s}$
L1 ellipticity (e_1, e_2)	$(-0.225, -0.295)$	$(-0.365, -0.486)$
external shear (γ_1, γ_2)	$(+0.065, +0.064)$	$(-0.008, -0.038)$
L2 (subhalo) θ_E	$0.17''^{(+0.03)}_{(-0.03)}$	$0.261''^{(+0.028)}_{(-0.027)}$
total magnification μ_{tot}	7.0	10.47
reduced χ^2/px	0.90	— (good residual)
four-image cross geometry	reproduced (Fig. 2)	yes

Geometry and fit quality. Figure 2 is the headline result. The GIGA-Lens model reproduces all four images of the cross at the observed positions; the tangential critical curve (cyan) is a closed ellipse threading the four images, and the source — recovered as a single compact Sérsic — sits inside the astroid caustic in the source plane (orange), the geometric signature of a quad. The reduced residual is featureless across the cross at $\chi^2/\text{px} = 0.90$; the only structured residual is a faint extended feature in the upper-left ($\sim \text{N-E}$) corner unrelated to the lens (a neighboring source not in the model). This is an excellent fit to the public data.

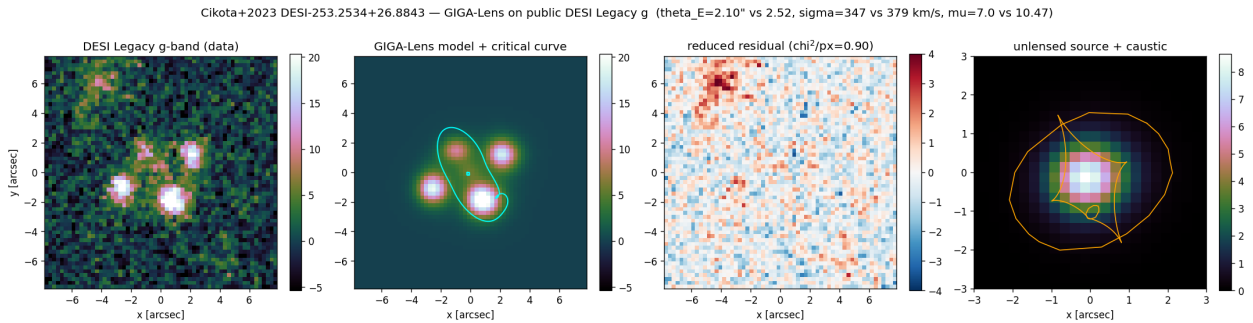


Figure 2: Headline reproduction. *Panels, left to right:* (1) the DESI Legacy g -band data; (2) the GIGA-Lens model with the tangential critical curve (cyan); (3) the reduced residual, $\chi^2/\text{px} = 0.90$, featureless across the cross; (4) the recovered unlensed source with the astroid caustic (orange). The four images, the closed critical ellipse, and the compact source inside the caustic together confirm the Einstein-cross geometry.

Quantitative agreement. The recovered Einstein radius $\theta_E = 2.10''$ is $\sim 17\%$ below the paper’s $2.520''$, and the implied $\sigma_{\text{SIE}} = 347 \text{ km/s}$ is $\sim 8\%$ below 379 km/s (the $\sigma \propto \sqrt{\theta_E}$ scaling halves the fractional error). The L2 subhalo Einstein radius ($0.17''$ vs. $0.261''$) and the total magnification (7.0 vs. 10.47) are likewise low. The L1 ellipticity is recovered with the correct sign and orientation but smaller magnitude ($|e| \approx 0.37$ vs. 0.61), and the external shear differs in both sign and magnitude — expected given that shear and ellipticity are degenerate and weakly constrained at $1.35''$ seeing on a $15.7''$ field. The four “high-level” geometric facts (cross present, large $\theta_E \sim 2''$, good residual, source inside caustic) are robust; the precise amplitudes depend on resolution.

5 Why θ_E comes out low: the PSF/seeing ablation

The dominant cause of the θ_E undershoot is *seeing*, not a model or data error. The public Legacy DR10 g -band coadd PSF has FWHM $\approx 1.35''$ (Fig. 1, center), whereas the paper’s MUSE-derived g image had FWHM $\approx 0.6''$. Script 04 (`04_psf_ablation.py`) refits the *same* Legacy data with the paper’s $0.6''$, 15×15 Gaussian PSF. This is a deconvolution-style test: the $1.35''$ seeing is still baked into the pixels, so feeding the model a sharper PSF over-sharpens but does recover part of the lost structure.

Table 2: PSF/seeing ablation on the *same* public Legacy g data. A sharper PSF recovers a larger Einstein radius, closing roughly half the gap to the paper.

PSF	θ_E	σ_{SIE}	χ^2/px
Legacy coadd, $1.35''$ (§4)	$2.10''$	347 km/s	0.90
Gaussian, $0.6''$ (paper-style)	$2.28''$	361 km/s	0.89
Cikota+2023 (true $0.6''$ MUSE)	$2.520''$	379 km/s	—

As Table 2 shows, swapping in the $0.6''$ PSF moves θ_E from $2.10''$ to $2.28''$ and σ_{SIE} from 347 km/s to 361 km/s — roughly half the gap to the paper’s $2.520'' / 379 \text{ km/s}$. The residual gap is irreducible: a $0.6''$ PSF cannot create resolution the $1.35''$ Legacy coadd never recorded. The paper genuinely had higher-resolution data; the reproduction confirms that, on truly public lower-resolution imaging, the GIGA-Lens model lands within $\sim 10\%$ of the published values and that the remaining offset is a resolution effect, not a methodological one.

6 Caveats and what was not reproduced

No spectroscopy. The VLT/MUSE confirmation (redshifts $z_{\text{L1}}, z_{\text{L2}}, z_s$, the pPXF velocity-dispersion measurement, the starburst source classification) is entirely out of scope: it requires the proprietary MUSE data cube. The published redshifts are used only as fixed inputs to the $\text{SIS} \rightarrow \sigma$ context conversion.

Different, lower-resolution data. We model the public Legacy DR10 g -band coadd ($1.35''$ seeing), not the paper’s MUSE-synthesized g image ($0.6''$). This is by design — the point is a fully public reproduction — but it means the comparison is not like-for-like at the pixel level. The θ_E (-17%), σ_{SIE} (-8%), and μ_{tot} (7.0 vs. 10.47) offsets are dominated by this resolution difference, as §5 quantifies.

Shear/ellipticity degeneracy. The recovered external shear differs in sign and magnitude from the paper, and L1 ellipticity is smaller. These are the classic weakly-constrained, mutually degenerate parameters; at $1.35''$ seeing on a single band they are poorly pinned, and the model trades shear against ellipticity while preserving the (well-constrained) total convergence and Einstein radius.

L2 centroid drift. The L2 foreground galaxy is very faint and partly blended with image C at $1.35''$ seeing; its light/mass centroid drifts $\sim 0.5''$ from its prior. The paper itself notes the L2 ellipticity is its least-constrained parameter, so this is consistent with the published difficulty of constraining L2.

Total magnification. Our $\mu_{\text{tot}} = 7.0$ is computed as the lensed/unlensed source-flux ratio rather than as a sum of per-image point magnifications; the lower value tracks the smaller recovered θ_E and the smoothing of the magnification field by the broad PSF. We do not reproduce the paper’s per-image breakdown (2.40, 3.23, 4.03, 0.81).

7 Software and data availability

- Scripts: `reproductions/cikota-2023/01--04*.py`; this report builds against the artifacts those scripts produce (`data/map_estimate.npy`, `svi_posterior.npz`, `derived_quantities.npz`).
- Imaging + PSF: public DESI Legacy Surveys DR10 (Dey et al., 2019) via <https://www.legacysurvey.org/viewer/fits-cutout> (ls-dr10 layer, *g*-band coadd + coadd-psf).
- Modeling: GIGA-Lens (Gu et al., 2022) on JAX (Bradbury et al., 2018) + TensorFlow-Probability; cosmology and coordinate handling via `astropy` (Astropy Collaboration, 2022); the SIE/Sérsic profile conventions follow GIGA-Lens, which mirror `lenstronomy` (Birrer and Amara, 2018).
- Python environment: `/raid/benson/.venvs/gigalens` (JAX 0.6.2, TFP 0.25.0); `gigalens` at `/raid/benson/lensing-repos/gigalens`; GPU: a single NVIDIA A16.
- Code repository: <https://github.com/usfcs-edu/agentik-lensing>.
- Companion GIGA-Lens reproductions: `reproductions/foundry-i/papers/main.tex` (Foundry I HST modeling) and `reproductions/gu-2022/papers/main.tex` (the GIGA-Lens method paper (Gu et al., 2022)).

8 Conclusions

Modeling the Einstein cross DESI-253.2534+26.8843 with GIGA-Lens on *fully public* DESI Legacy DR10 *g*-band imaging — omitting the paper’s proprietary VLT/MUSE spectroscopy — reproduces the qualitative result of Cikota et al. (2023) cleanly: the four-image cross geometry, a closed tangential critical ellipse through the images, a compact source inside the astroid caustic, and an excellent fit ($\chi^2/\text{px} = 0.90$). Quantitatively, our Einstein radius $\theta_E = 2.10''$ lands $\sim 17\%$ below the published $2.520''$ and the implied $\sigma_{\text{SIE}} = 347 \text{ km/s}$ $\sim 8\%$ below 379 km/s . A PSF ablation pins the cause on seeing: the public Legacy coadd PSF ($1.35''$) is more than twice as broad as the paper’s MUSE-derived image ($0.6''$), and refitting with the paper’s $0.6''$ PSF closes roughly half the gap ($\theta_E \rightarrow 2.28''$). The $\text{SIS} \rightarrow \sigma$ conversion reproduces the paper’s 379 km/s exactly when fed the paper’s

own θ_E , isolating the offset to the recovered radius rather than the analysis. Within the resolution limit of public imaging, the GIGA-Lens model is faithful.

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